Core Concepts of Generative AI for Developers

The Revolution Created by Generative AI

Xavier Morera

Helping developers create epic generative applications

www.xaviermorera.com / www.lupo.ai

Generative AI is changing how individuals work, what they create, and at the speed at which everything is created.

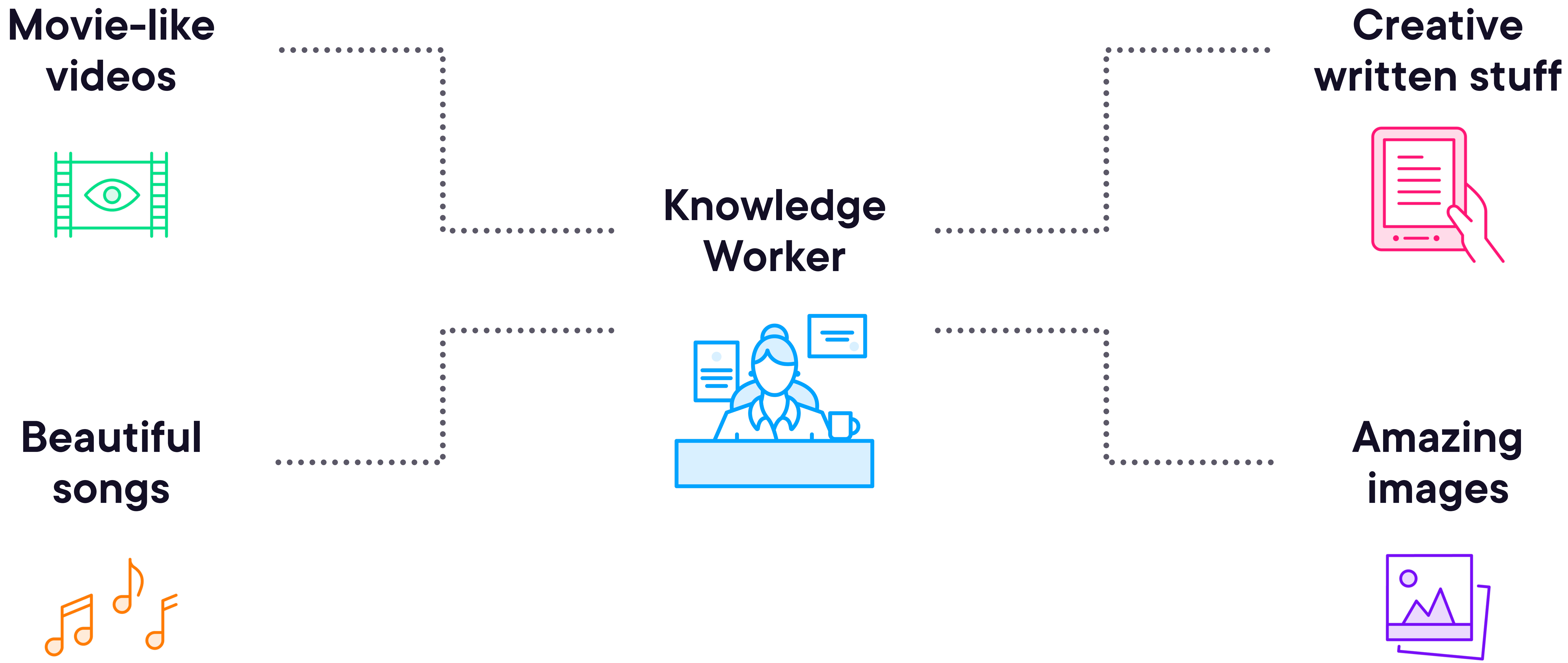
Enterprises need to adapt and embrace AI to stay relevant and prosper.

How Generative AI Changed the World?

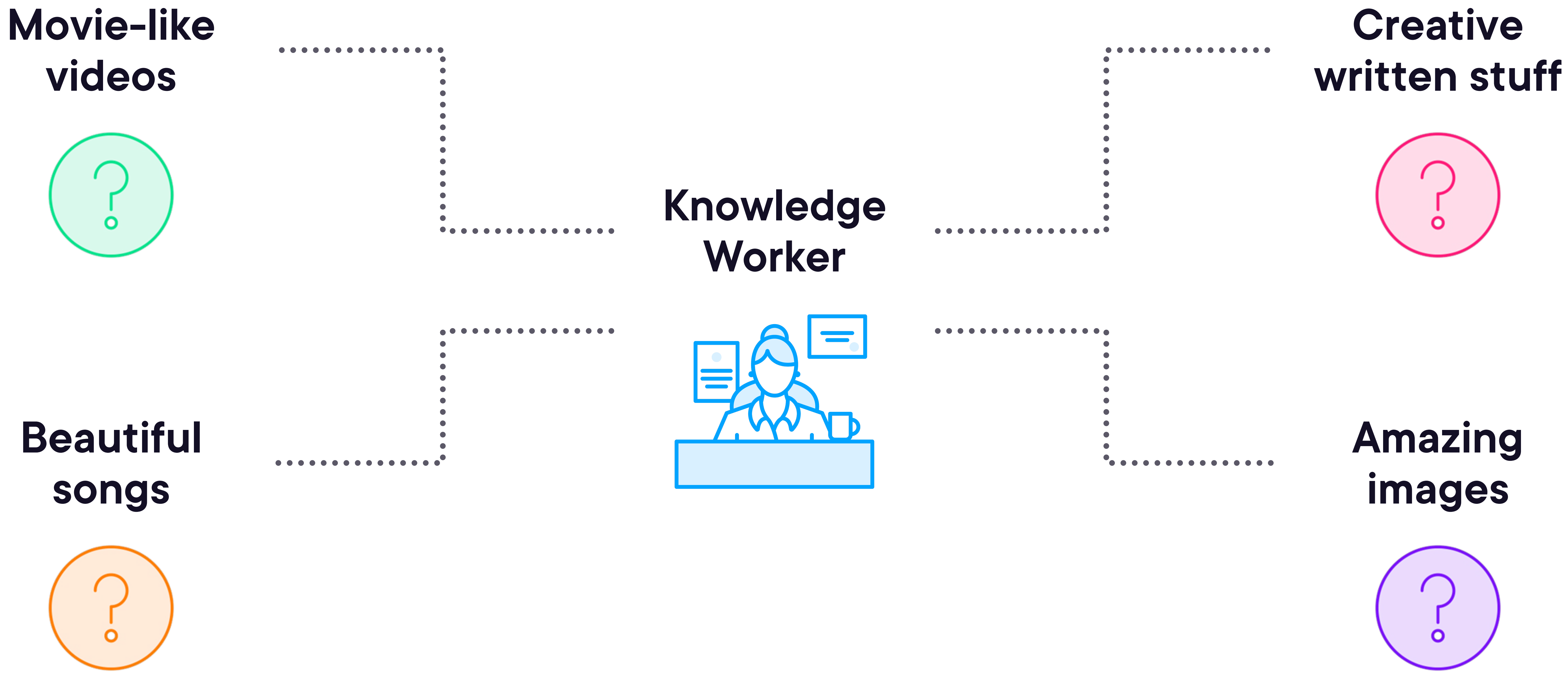
"AI will not replace you, but someone using AI will replace you".

** not entirely true... but let's keep this thought for a second...*

With Generative AI



With Generative AI



ChatGPT 4

You

Write a blog post on innovation in the medical industry.

ChatGPT

Innovation in the medical industry has always been at the forefront of pushing the boundaries of what is possible in healthcare and treatment. As technology evolves, so too does the capacity for groundbreaking advancements that can significantly improve patient outcomes, enhance efficiency in healthcare delivery, and reduce costs. This blog post explores several key areas where innovation is reshaping the medical industry, including digital health, personalized medicine, robotics, and artificial intelligence (AI).

Digital Health: Bridging the Gap

Digital health is revolutionizing the way healthcare is delivered. Telemedicine, for example, has become increasingly popular, especially in the wake of the COVID-19 pandemic, allowing patients to consult with healthcare professionals remotely. This not only makes healthcare more accessible but also reduces the strain on healthcare facilities. Mobile health apps and wearable devices are providing patients with more control over their health and well-being, enabling real-time monitoring of vital signs, medication management, and fitness tracking.

Personalized Medicine: Tailored Treatments

Message ChatGPT...

?

ChatGPT 4

ChatGPT

Here's the image of a cyclist going uphill in the mountains of Italy. The scene captures the essence of a challenging yet picturesque bike ride through the Italian countryside.

Message ChatGPT...

ChatGPT can make mistakes. Consider checking important information.


Midjourney
newbies-156
Bot room for new users. Type /imagine then describe what you want to draw. See https://docs.midjourney....
Search
6 Events
6
Channels & Roles
NEW
INFO
getting-started
announcements
status
rules
community-updates
micro-polls
SUPPORT
trial-support
support
NEWCOMER ROOMS 3
newbies-126
NEWCOMER ROOMS 4
newbies-156
newbies-186
CHAT
xmorera
Open in Browser
Message #newbies-156
GIF

SOUNDRAW

Create Music License FAQ API for Businesses Pricing Log In Sign Up

Genre Mood Theme Length 3:00 Tempo Instruments

+ Video Preview

Hip Hop Trap Drill R&B Jersey Club Latin Acoustic Rock Ambient Beats Drum n Bass Electro & Dance Funk House

Techno & Trance Tokyo night pop Pop Lofi Hip Hop World Electronica Orchestra Tropical House Afrobeats Phonk Christmas

Hip Hop ✕

001
Hip Hop

Dark, Laid Back, Hopeful, Sentimental,... 100 BPM 3:04

0:00 0:09 0:19 0:28 0:38 0:48 0:57 1:07 1:16 1:26 1:36

Mixer	Mid	Mid	Extreme	Extreme	Mid	Mid	Quiet	Quiet	Mid	Mid	Extreme
Melody											
Backing											
Bass											
Drum											

Prompt: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots,...

more

0:03 / 0:59

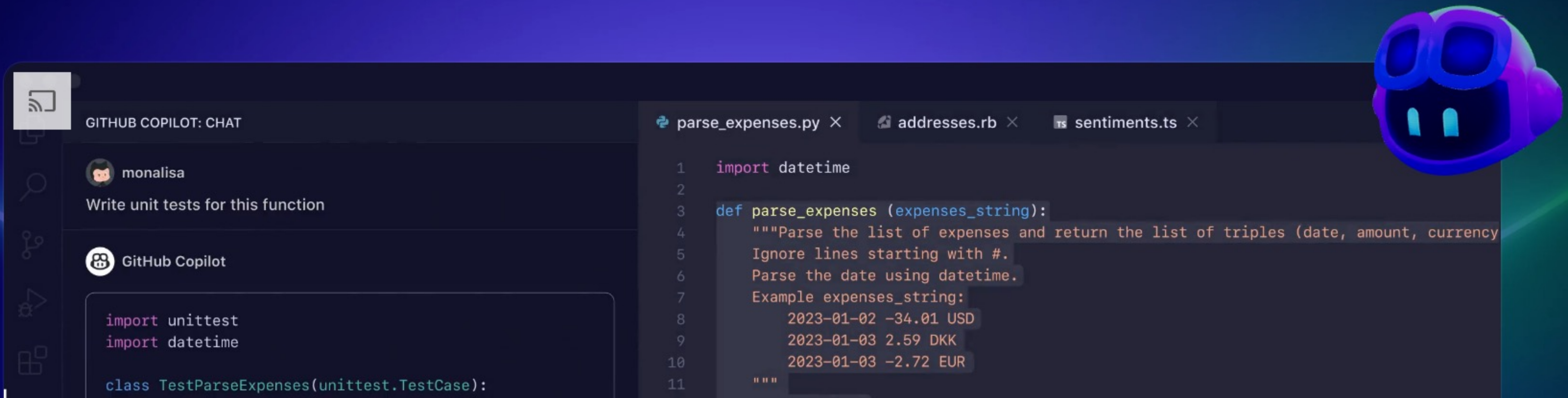
Prompt: Several giant w
long wooly fur lightly bl

more

GitHub Copilot

The world's most widely adopted AI developer tool.

Get started with Copilot >



ChatGPT 3.5

You

Create a function that parses a json file and returns repeated values

ChatGPT

Certainly! Below is a Python function that reads a JSON file, parses it, and returns a dictionary containing the repeated values along with their counts:

python   Copy code

```python
import json
from collections import Counter

def find_repeated_values(json_file):
    with open(json_file, 'r') as f:
        data = json.load(f)

    # Flatten the JSON data to a single list
    flattened_data = []

    def flatten_json(json_data):
        if isinstance(json_data, dic
            for key, value in json_data.items():
```

Message ChatGPT...

ChatGPT can make mistakes. Consider checking important information.

Synthetic Data. Better than real.

Still struggling with real data? Use existing data for synthetic data generation. Synthetic data is more accessible, more flexible, and simply...smarter.

Request a demo

100% compliance?

citi City of Vienna ERSTE merkur VERSICHERUNG innovation lab Telefónica

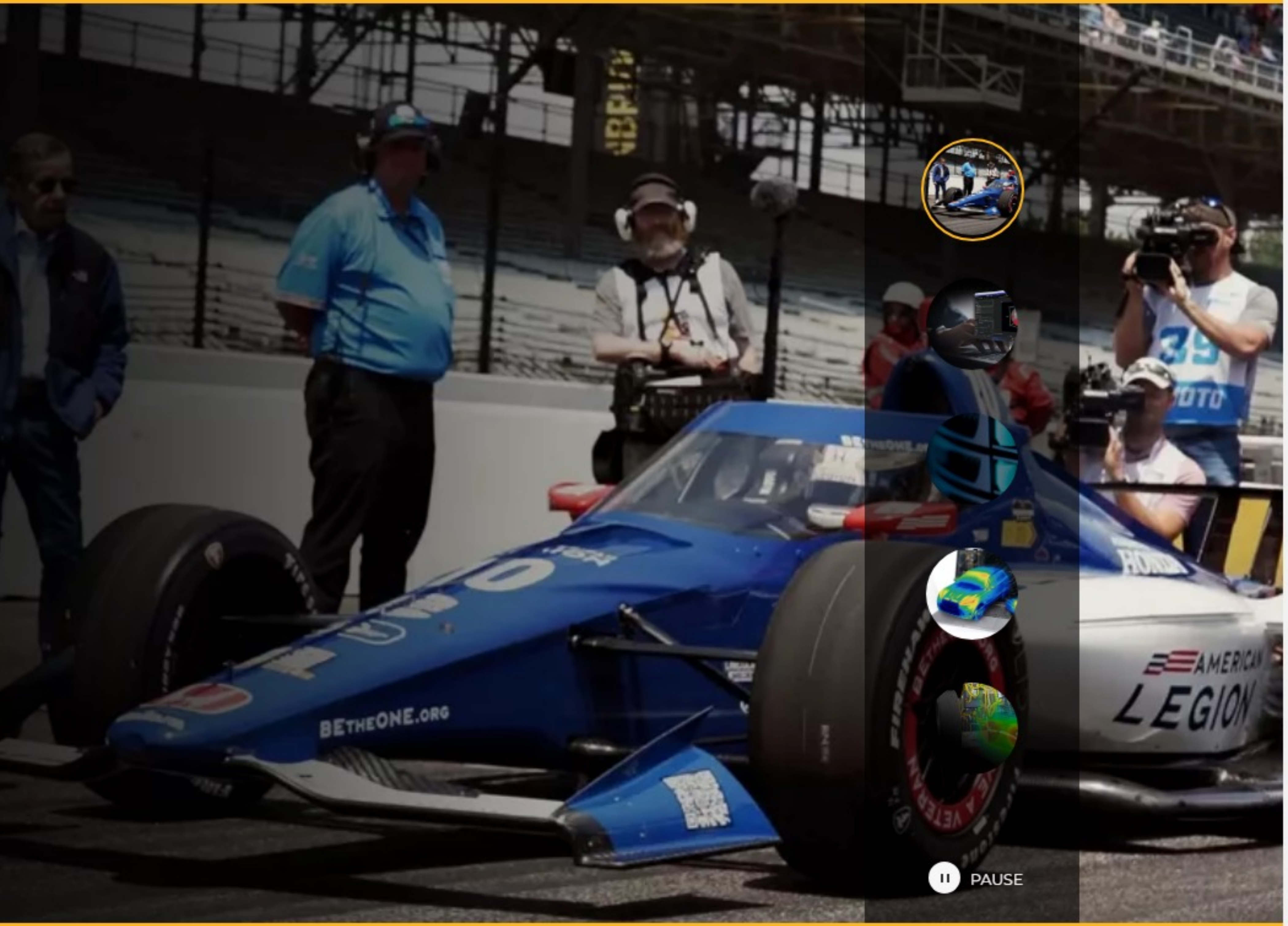
What is synthetic data, anyway?

POWERING INNOVATION THAT DRIVES HUMAN ADVANCEMENT

Honda Makes IndyCar History

See how Ansys simulation technology gives Honda Racing Corporation USA engineers the ability to confidently replicate track conditions without needing physical testing – getting valuable data in a fraction of the time.

WATCH EPISODE 7



PAUSE

SEE WHAT ANSYS CAN DO FOR YOU

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Learn the impact of AI on tech job security and strategies for technologists to future- careers with AI training and new technology adoption.

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← Back to Help Home

How Netflix's Recommendations System Works

Our business is a subscription service model that offers personalized recommendations, to help you find shows and movies of interest to you. To do this we have created a proprietary, complex recommendations system. This article provides a high level description of our recommendations system in plain language.

Related Articles
Plans and Pricing
How to sign up for Netflix
What is Netflix?
Getting started with Netflix
How do I create a Netflix account for someone else?

The basics

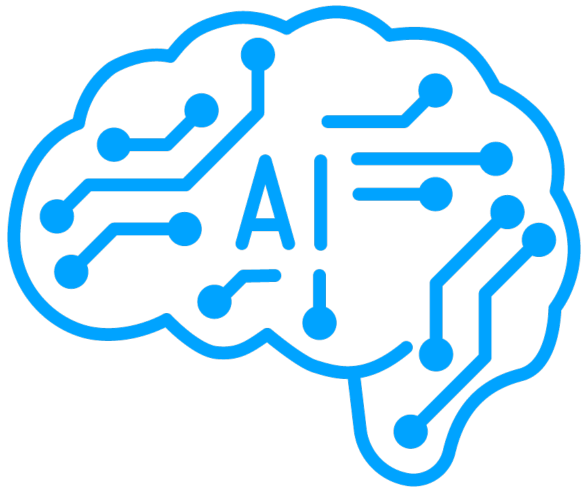
Whenever you access the Netflix service, our recommendations system strives to help you find a show or movie to enjoy with minimal effort. We estimate the likelihood that you will watch a particular title in our catalog based on a number of factors including:

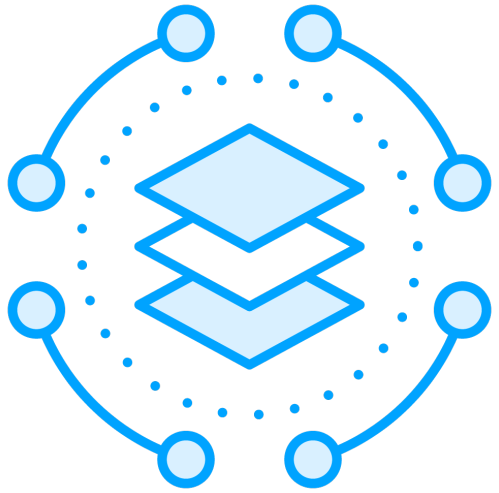
- your interactions with our service (such as your viewing history and how you rated other titles),
- other members with similar tastes and preferences on our service, and
- information about the titles, such as their genre, categories, actors, release year, etc.

In addition to knowing what you have watched on Netflix, to best personalize the recommendations we also look at things like:

- the time of day you watch,
- the devices you are watching Netflix on, and

Generative AI for Product Development





AI is reshaping the product development landscape through generative design

Contributes to material optimization

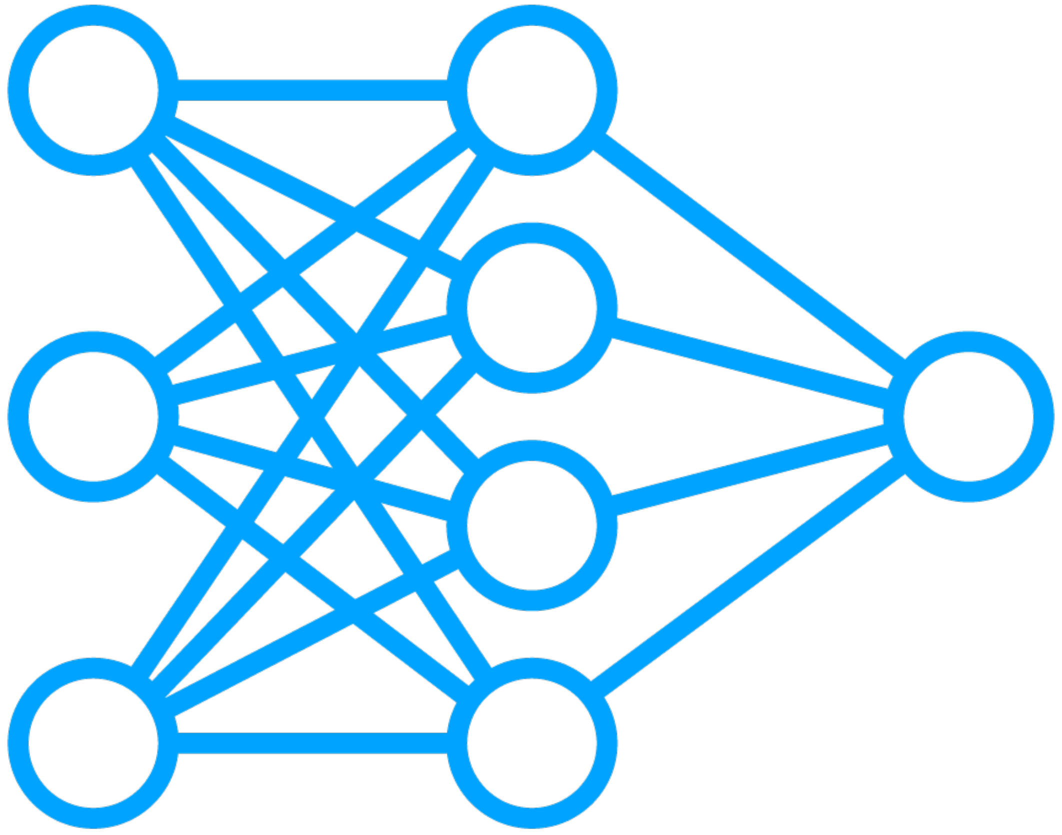
Improves maintenance through predictive analysis

Also used to design new drugs, material, and proteins

Businesses are undergoing a transformative shift with Generative AI.

Enablers of Generative AI

Technological Advances

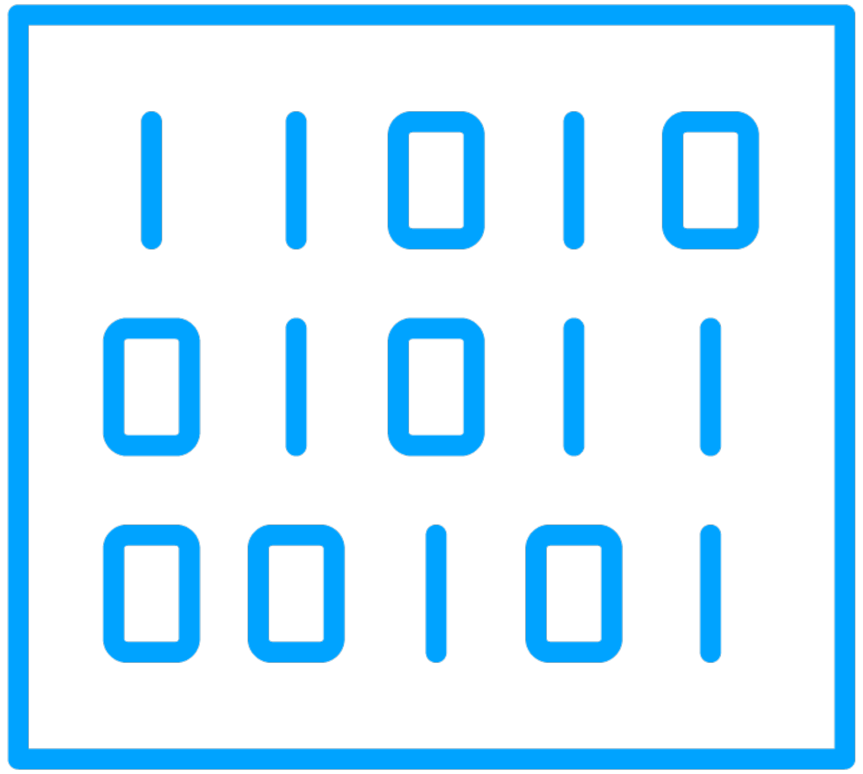


Breakthroughs in neural network architectures have improved Gen AI model capabilities

- Especially transformers

Enable generation of sophisticated text, images, video, and music

Data Availability



Digital age resulted in an explosion of data

Gen AI models thrive on large datasets

- For training purposes
- Abundance of data facilitated rapid progress

Computational Power

Increases in computational power have made it possible to train large and complex models

- Specialized hardware
- Like GPUs and TPUs

At scales not possible before

Versatility and Application

Gen AI has a wide range of applications

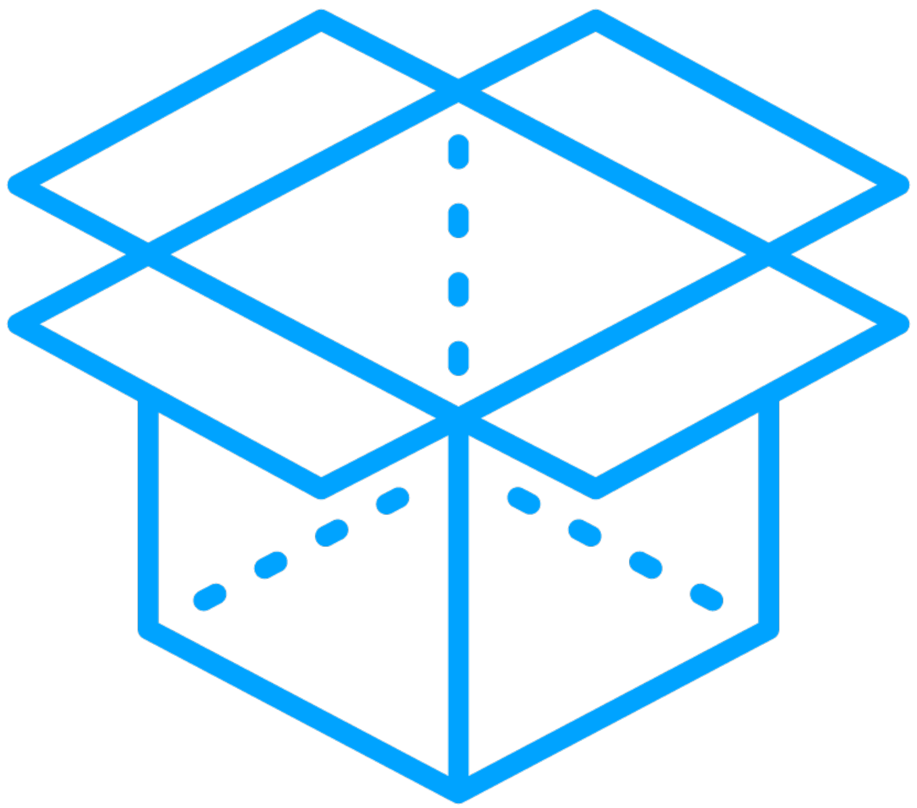
Content creation, solving complex problems, automating tasks…

Investment and Interest

Potential of Gen AI has attracted a significant invest from private and public sectors

Financial backing supports research, development, and deployment of Gen AI tech

Collaboration and Open Source

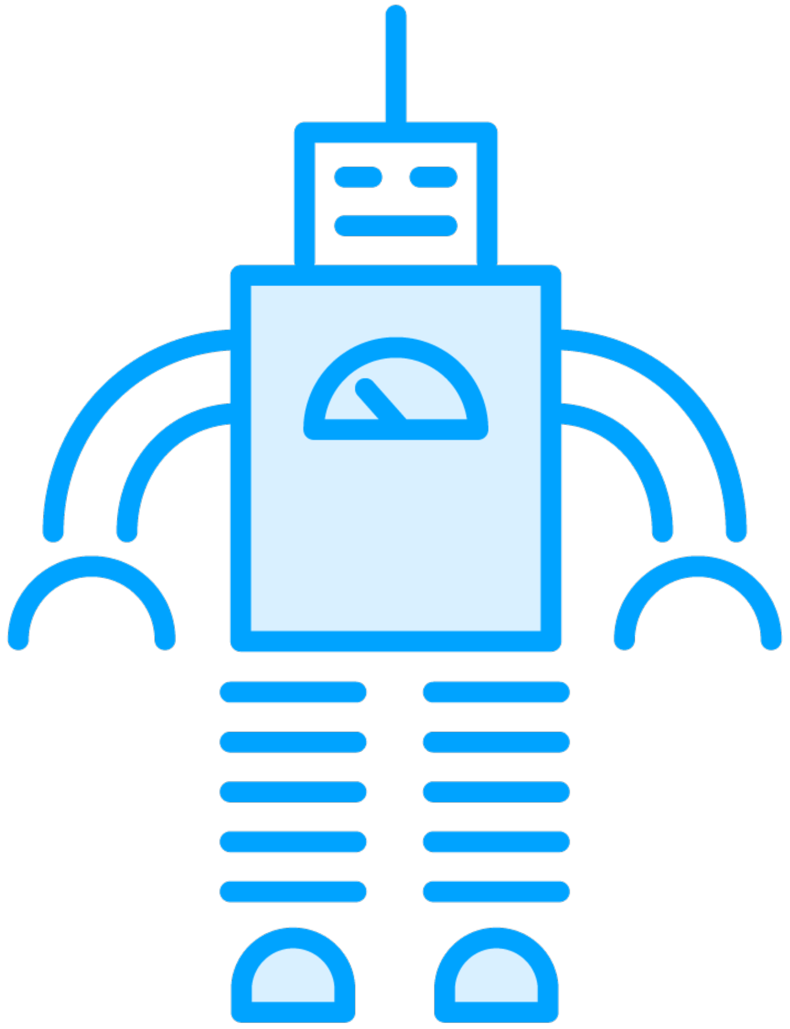


Gen AI benefits from

- A culture of collaboration
- Open-source sharing of models and techniques

Accelerates innovation and application of Gen AI technologies

Economic and Efficiency Gains



Gen AI can automate tasks that were done traditionally by humans

- Leading to efficiency gains

Opens new business models and opportunities

- Economically attractive

These factors have created a fertile ground for rapid growth and innovation.

Makes Gen AI one of the most exciting and dynamic areas of technology today.

Where Did Generative AI Come From? A Few Brief History Points

Where Did Gen AI Come From?

1960s
Eliza
First chatbot

1974-1980 and 1978-2000
AI winter
Periods of reduced funding and interest

2006
Deep learning breakthroughs
New techniques developed

2014
Introduction of GANS
Accelerated research

2017
Transformer architecture
Attention is all You Need paper

2022
ChatGPT released
Public awareness

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A Few Important Things on How AI Works

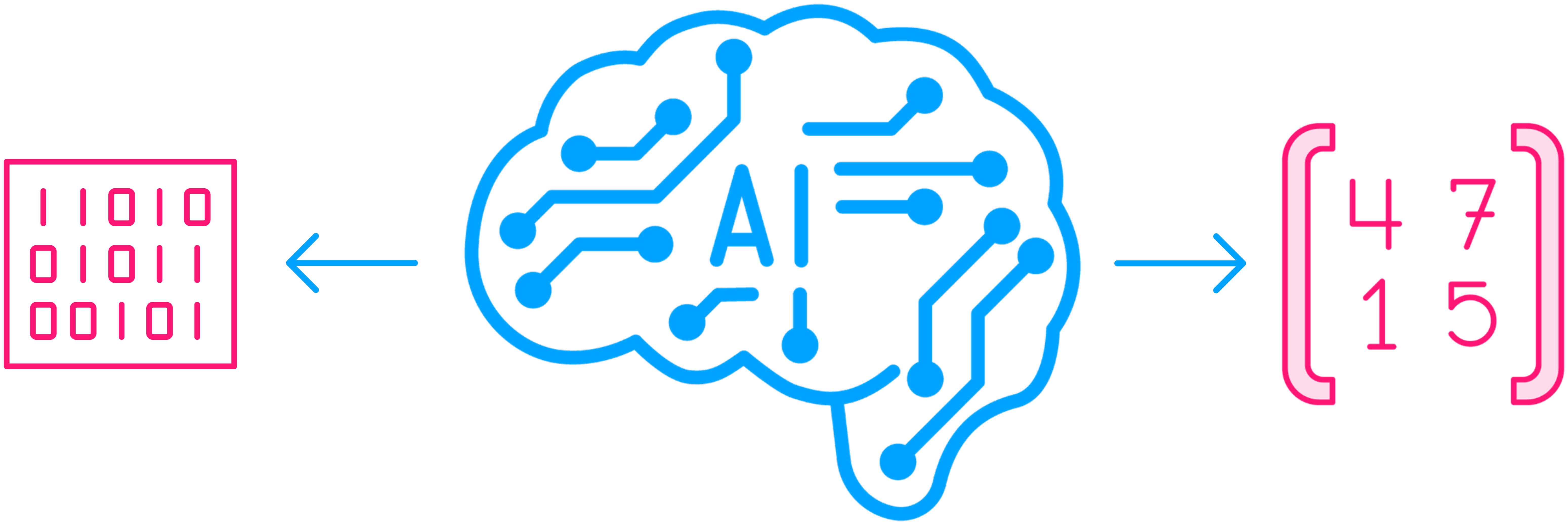
How Traditional Programming Works

vs

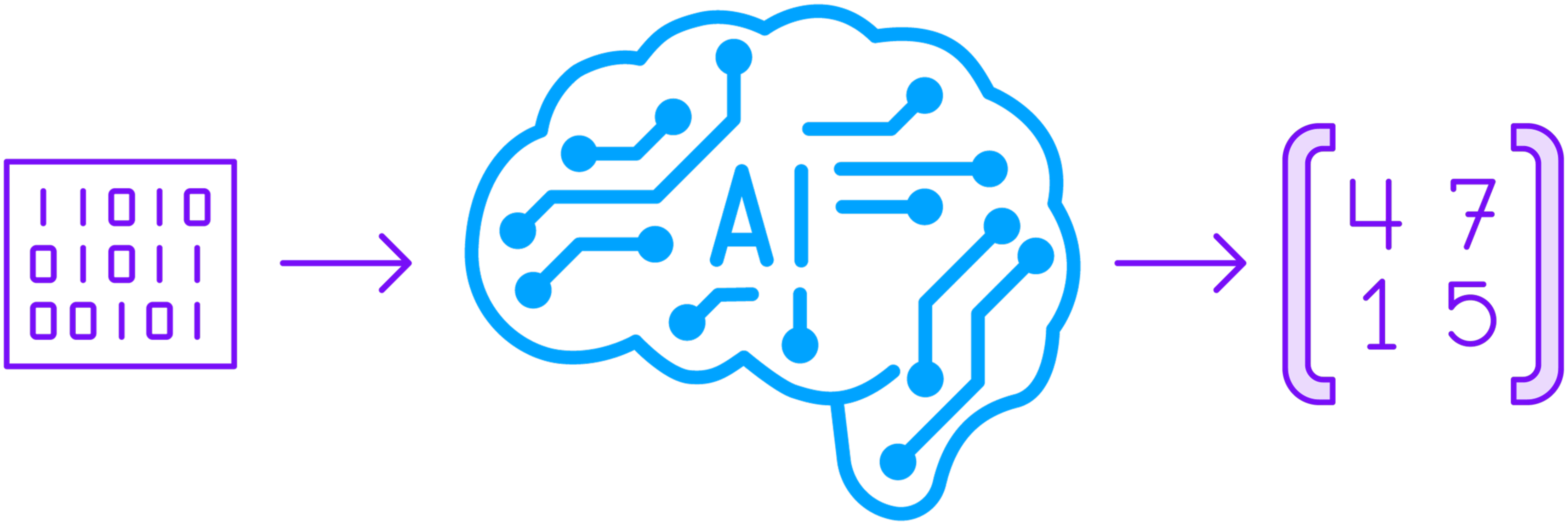
How AI Works

How Traditional Programming Works

How AI Works



How AI Works



Machine Learning Process

Define the problem

Collect the data and get it ready
(Prepare, clean, wrangle, normalize, feature selection...)

Select the right model for the job

Train the model
(What I just mentioned a few moments ago)

Evaluate output, fix, improve, and iterate until desired results

Machine Learning with Python - Practical Application

by Xavier Morera

Many problems are solved using Machine Learning. This course will teach you how to pick the ML algorithm that can help you create the right ML model to solve the problem at hand.

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Course author

Xavier Morera

Xavier is very passionate about teaching, helping others understand search and Big Data. He is also an entrepreneur, project manager, technical author, trainer, and holds a few certifications with...

Course info

Level	Beginner
Rating	(14)
My rating	
Duration	1h 56m
Released	21 Dec 2021
Reviewed	31 Oct 2022

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Types of Models Available

- Transformers
- Variational Autoencoders
- Generative Adversarial Networks
- Autoregressive
- Diffusion
- Many more...

Up Next:

Generative AI Categories: Text, Image, Video, and Audio